

Statistics for Data Insights



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With assistance from AI tools, particularly
Microsoft Copilot and Google Gemini

Outline

- Introduction to Data Analytics
 - Question 1 - What type of data are you working with?
 - Question 2 - What do you want to communicate?
 - Question 3 — Who is the audience and what do they need?
- 1. Measures of Central Tendency
- 2. Measures of Dispersion
- 3. Measures of Distribution Shape: Skewness and Kurtosis
- 4. Analysis of Relationships Between Variables

Introduction to Data Analytics

What is data analytics?

Data analytics = ask a clear question → let the data answer it → say what to do next.

Four questions data can answer: describe → explain → predict → decide — and where each one is covered in this course

Descriptive

✓ OUR FOCUS — 2 LECTURES

"What happened?"

 COVERED BY

- **Statistics for Data Insights**

Summaries, distributions, comparing groups.

- **Data Visualization**

Chart choice, plotting and data storytelling.

Diagnostic

LATER IN THE COURSE

"Why did it happen?"

 COVERED BY

- **Probability, Sampling & Uncertainty**
Select a sample and draw reliable conclusions about the population.
- **Testing, Estimation & A/B Testing**
Is the observed difference real or random variation (noise)?
- **Causal Inference & Experiments**
Determine whether an action caused the outcome.

Predictive

LATER IN THE COURSE

"What is likely next?"

 COVERED BY

- **Regression & Classification**
Predict a number or a category.
- **Clustering & Dimensionality Reduction**
Find hidden groups and structure.

Prescriptive

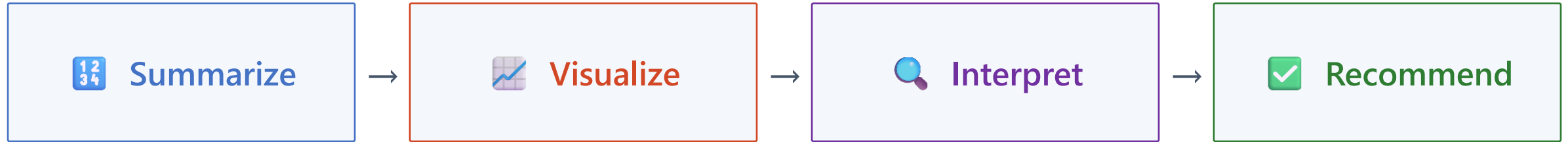
NOT COVERED

"What should we do?"

- Optimization, simulation, decision models that recommend the best action.

How we cover descriptive analytics: two lectures 🌐

One workflow, split in two — first find the story in the numbers, then show it so other people act on it



LECTURE 1

Statistics for Data Insights

Discover what the numbers are really telling you and find the story behind the numbers

-  Summary statistics - mean vs median, spread
-  Distributions - shape, skew, outliers
-  Comparing groups and relationships
-  From a raw number to a stated insight

LECTURE 2

Data Visualization

Visualize data to reveal insights for decision-making

-  Choosing a chart by purpose and audience
-  Building charts with Seaborn and Plotly
-  Storytelling - headline, evidence, action

3 key questions before any analysis 🧭

1

What **type of data** are you working with?

Categorical? Numeric? Date-time? Geospatial? Hierarchical?

 **STATISTICS** Counts and proportions for categories; mean, median and spread for numbers.

 **VISUALIZATION** Bar or stacked bar for categories; histogram or line for numbers over time.

2

What do you want to **communicate**?

Comparison? Composition? Relationship? Distribution? Trend?

 **STATISTICS** **Group difference** → compare means; **relationship** → correlation; **spread** → quartiles.

 **VISUALIZATION** One chart, one message — the statistic you quote is the chart you draw.

3

Who is the **audience** and what do they need?

Analyst? Manager? Executive? General public?

 **STATISTICS** Analysts want the full distribution; executives want one number in plain words.

 **VISUALIZATION** Match the detail to the decision — same finding, different chart and depth.

Question 1 - What type of data are you working with?

Understanding the variable types is crucial for selecting appropriate statistical methods and visualization techniques



Question 1a — Variable types: what each column is

Five column types. Identify the column type first — it determines which statistics and which charts are appropriate.

Data type	Description & examples	Recommended statistics	Recommended visualizations
 Categorical - Nominal	Categories without a natural order — region, payment method, product category.	Counts, proportions, mode	Bar chart, pie chart
 Categorical - Ordinal	Categories with a meaningful order — Low/Medium/High, star ratings.	Counts, proportions, median, percentiles	Ordered bar chart
 Numeric	Measurable quantities — revenue, units sold, age.	Mean, median, standard deviation, range	Histogram, box plot, scatter plot
 Date-time	Time-based data points — order date, month, week.	Growth rate, period-over-period change	Line chart, time-series plot
 Boolean	Binary yes/no or true/false values — returned?, churned?, first-time buyer?	Rates, proportions	100% stacked bar, donut chart

Question 1b — Data shapes: how the table is organised 📁

The same variables can answer different questions: how they are arranged determines the analysis and the chart family.

 **Cross-sectional** One snapshot, many entities — sales by region this month. → Compare and rank; bar charts.

 **Time-series** The same measure repeated over time. → Growth and trend; line or area charts.

 **Geospatial** Country, region, postcode, latitude-longitude. → Regional rates; choropleth or bubble map.

 **Hierarchical** A part inside a whole — category → sub-category → product. → Share of total; treemap or stacked bar.

 **Financial** Open-high-low-close, budget versus actual. → Variance; candlestick or bullet chart.

 **Relational** Links between entities — who buys with whom. → Co-occurrence; network graph or heatmap.

Categorical Data

Qualitative (non-numeric) data that represents groups or categories

Nominal

Categories with no inherent order

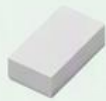
Examples



Pen



Pencil



Eraser



Cow



Dog



Cat

Characteristics

- Distinct categories
- No natural order
- Labels or names (e.g., product type, country, animal)

Common charts



Bar chart



Pie chart



Count plot

Ordinal

Categories with a meaningful order

Examples



Excellent



Good



Okay



Bad



Very bad



Strongly disagree



Disagree



Neutral



Agree

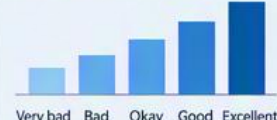


Strongly agree

Characteristics

- Ordered categories (ranking)
- Can be sorted from low to high
- Common for ratings, levels, and preferences (e.g., satisfaction, education level, agreement)

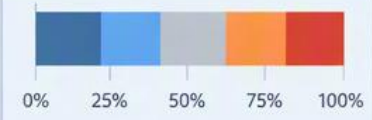
Common charts



Ordered bar chart



Stacked bar chart



Likert chart (100% stacked)



Categorical data describes groups or labels, not numbers.

Identify whether the data is nominal or ordinal to choose the right analysis and visualization.

Numerical Data

Quantitative (numeric) data that represents counts or measurements

Discrete

Numbers obtained by counting

Examples



Number of students



Number of orders



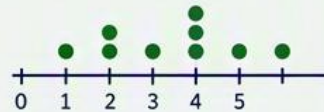
Number of cars



Number of defects/items

Key characteristics

- Countable values
- Usually whole numbers (integers)
- Gaps between possible values
- Cannot have fractions (e.g., 2.5 students or 3.7 cars)



Only specific values
(e.g., 0, 1, 2, 3, ...)

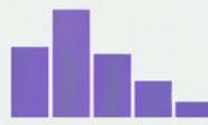
Common charts



Bar chart
(counts)



Dot plot



Count plot

Continuous

Numbers obtained by measuring

Examples



Height
(175.3 cm)



Weight
(72.5 kg)



Temperature
(36.8 °C)



Distance
(12.7 km)

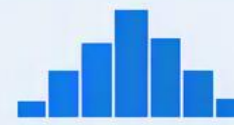
Key characteristics

- Measured values
- Can contain any real value (decimals)
- Infinite possible values within a range
- Precision depends on measurement

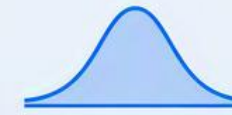


Any value within a range
(e.g., 12.7, 18.4, 25.0, ...)

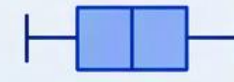
Common charts



Histogram



Density plot



Box plot



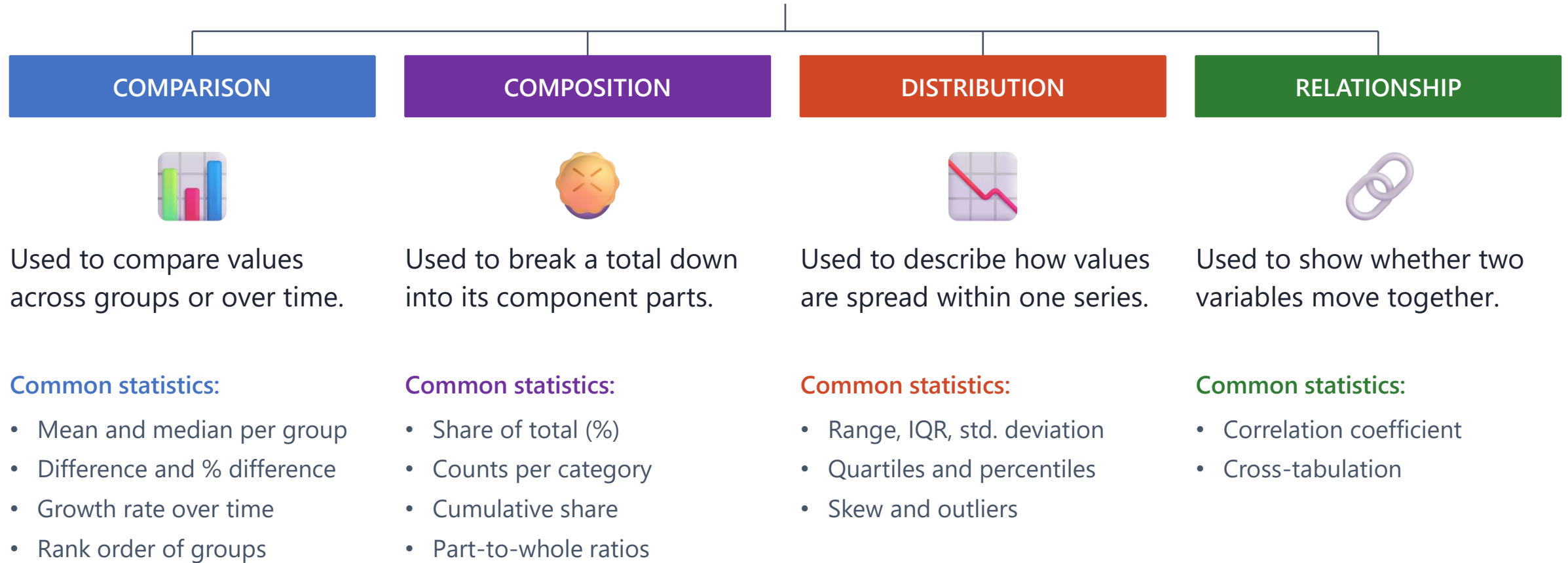
Numerical data describes quantities and amounts.

Identify whether the data is discrete or continuous to choose the right analysis and visualization.

Question 2 - What do you want to communicate?

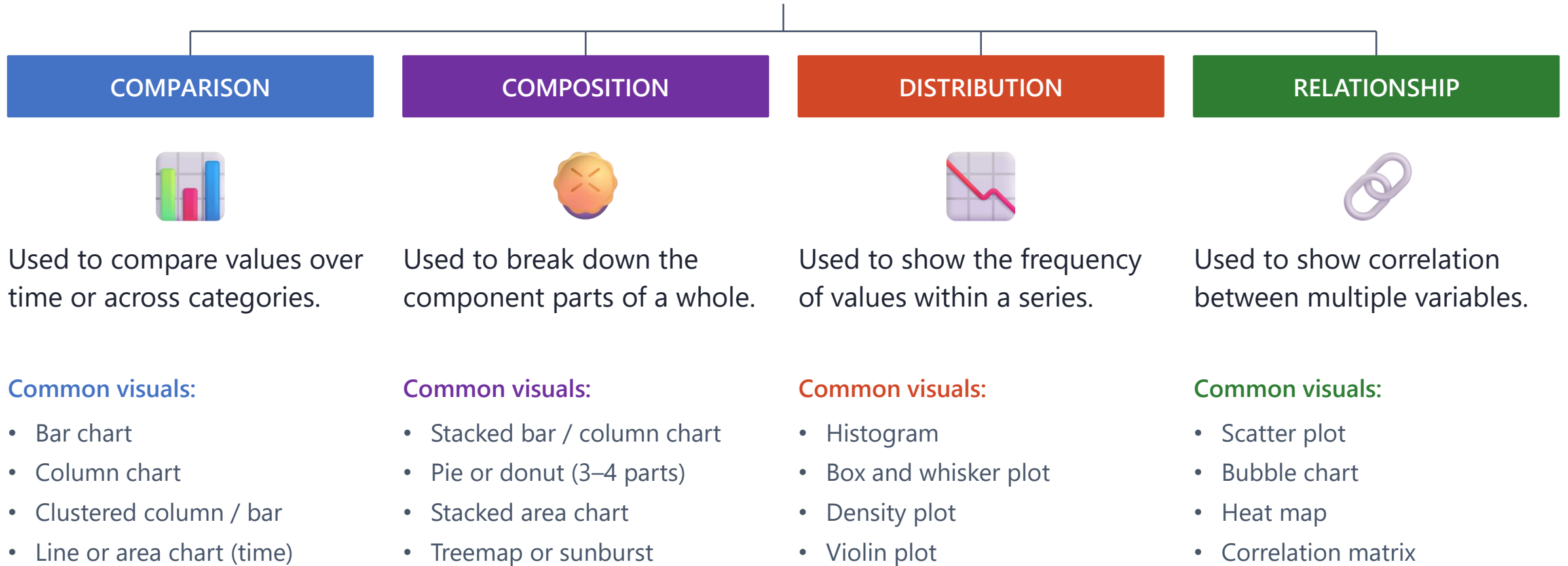


2 What do you want to **communicate**? — the statistics



 **Start simple:** the mean, median, a percentage and a growth rate answer most business questions. Use fancier measures only when these fall short.

② What do you want to **communicate**? — the charts



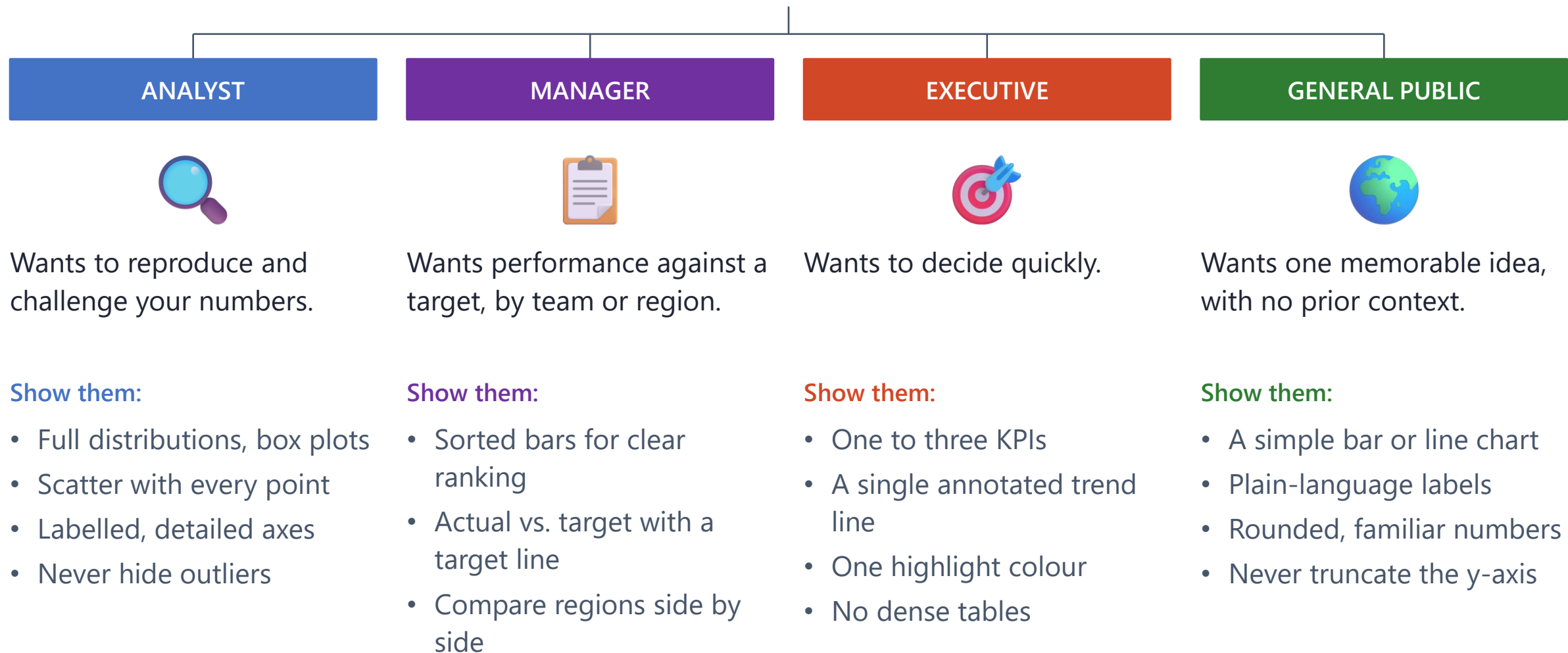
💡 **Keep it simple!** Basic variations of bar, column and line charts are often the simplest — and clearest — way to visualize data.

**Question 3 — Who is the audience
and what do they need? 🧑🧑**



3

Who is the **audience** and what do they need?



Ask before you build: what decision will this person make, what do they already know, and how many seconds will they really spend looking? Those three answers set the level of detail.

Statistics for Data Insights

Describing data with numbers: four topics, one workflow

1

Central Tendency

What single value is typical of the data?

2

Dispersion

How far do values spread around that centre?

3

Distribution Shape

Is the data symmetric, skewed or heavy-tailed?

4

Relationships

Do two variables move together?

Work in order: a centre is only meaningful once you state its spread, its shape, and the sample size n .

Learning outcomes

- Compute and interpret the mean, median, mode, range, variance, standard deviation, percentiles and IQR.
- Diagnose distribution shape with skewness and kurtosis, and detect outliers.
- Measure and correctly report the relationship between two variables.

1

Measures of Central Tendency

You will learn to choose and defend the single number that best represents a dataset.

What “Typical” Means

Three definitions of the centre — the question decides which one is correct

$\bar{x}$

Mean

The balance point of the data. Uses every value, so extreme values move it.

Best for: symmetric data, and any figure that must multiply up to a total.

$$\bar{x} = \sum x_i / n$$

M

Median

The middle value once the data is ordered. Uses position only, not magnitude.

Best for: skewed data, or data containing outliers.

M = middle of ordered data

Mo

Mode

The most frequently occurring value. The only centre defined for categories.

Best for: categorical data and repeated discrete values.

Mo = most frequent value

All three answer “what is typical?” — they disagree whenever the distribution is not symmetric.

In a perfectly symmetric distribution mean = median = mode. The size of the gap between them is itself a measurement of asymmetry.

Worked Example: Five Trainees

Small enough to verify every statistic by hand

Name	Gender	Speaking /50	Math /50
Omar	M	40	38
Layla	F	45	41
Noura	F	23	42
Khalid	M	39	48
Yousef	M	39	32

Ordered speaking marks

23 • 39 • 39 • 40 • 45

Interpretation: a typical speaking mark is 39; the mean of 37.2 is dragged down by one low score.

Mean = 37.2

$(40 + 45 + 23 + 39 + 39) \div 5 = 37.2$ marks. Noura's 23 pulls it below every other mark.

Median = 39.0

Third value of five once ordered. Unaffected by how low the lowest mark is.

Mode = 39

39 occurs twice; every other mark occurs once. The most common performance.

Mean vs Median: The Effect of One Outlier

A controlled demonstration of what “robust” actually buys you

Before — clean data

Speaking: 40, 45, 23, 39, 39

Mean 37.2 • Median 39.0

Gap = 4.6% of the median — either centre is defensible.

After — one typing error (45 → 450)

Speaking: 40, 450, 23, 39, 39

Mean 118.2 • Median 40.0

Gap = 195% — the mean is now a mark nobody scored.

Statistic	Before	After	Change	Why
Mean	37.2	118.2	+218%	Uses the magnitude of every value
Median	39.0	40.0	+2.6%	Uses rank only; one value moves it one position

Practical rule: compute both on every numeric column. Investigate any gap larger than 10% of the median.

The Mode and Categorical Data

Counts and proportions, not averages

Payment method recorded for 200 orders

Value	Frequency	Proportion
Card	96	0.48
Cash	62	0.31
Wallet	30	0.15
Transfer	12	0.06
Total	200	1.00

Frequency and proportion

Frequency = number of times a value occurs. Proportion = frequency $\div$ sample size n .

Proportions always sum to 1 (100%).

Mode = “Card”

The most frequent level — the only measure of central tendency valid for categorical data.

Watch out

If all levels occur equally often there is no useful mode. On continuous data, group values into bins first.

Choosing the Right Centre

Measure the gap, then match the centre to the question

$$\text{Gap} = |\bar{x} - M| / M$$

A two-second test you can run on every numeric column

Gap	Skewness	What to report
< 5%	< 0.5	Mean alone is sufficient
5 – 20%	0.5 – 1	Report both; name the one used
> 20%	> 1	Median, plus the 90th percentile
any, n < 30	—	Raw values, not a centre

Match the centre to the question

“What does a typical order look like?”

Median — you are describing one unit.

“What revenue will 10,000 orders give?”

Mean — only the mean multiplies up to a total.

“Which box size do we stock most?”

Mode — a category question needs the most frequent level.

“What can we promise customers?”

Neither — use the 95th percentile.

Report it fully: “Median order value 268 QAR (n = 2,400; mean 418.60, right-skewed).”

Central Tendency in Pandas

Three methods, plus the split that makes them honest

PYTHON — CENTRE

```
df['speaking'].mean()      # 37.2
df['speaking'].median()    # 39.0
df['speaking'].mode()[0]   # 39
```

```
df.groupby('gender')['speaking'].mean()
# F  34.00   |   M  39.33
```

PYTHON — ONE-LINE SUMMARY

```
df[['speaking', 'math']].describe()
# count  mean   std   min   max
#   5.00  37.20  8.32  23.0  45.0
```

How to read the output

Row	Reading
count	5 of 5 — the column is complete. Check this first.
mean	37.20 — the balance point of all five marks.
50%	39.00 — the median; compare it with the mean.
std	8.32 — average distance from the mean, in marks.
min / max	23 and 45 — the two values driving the range.

Always split before you summarise: one overall mean of 37.2 hides a 5.3-mark gap between groups.

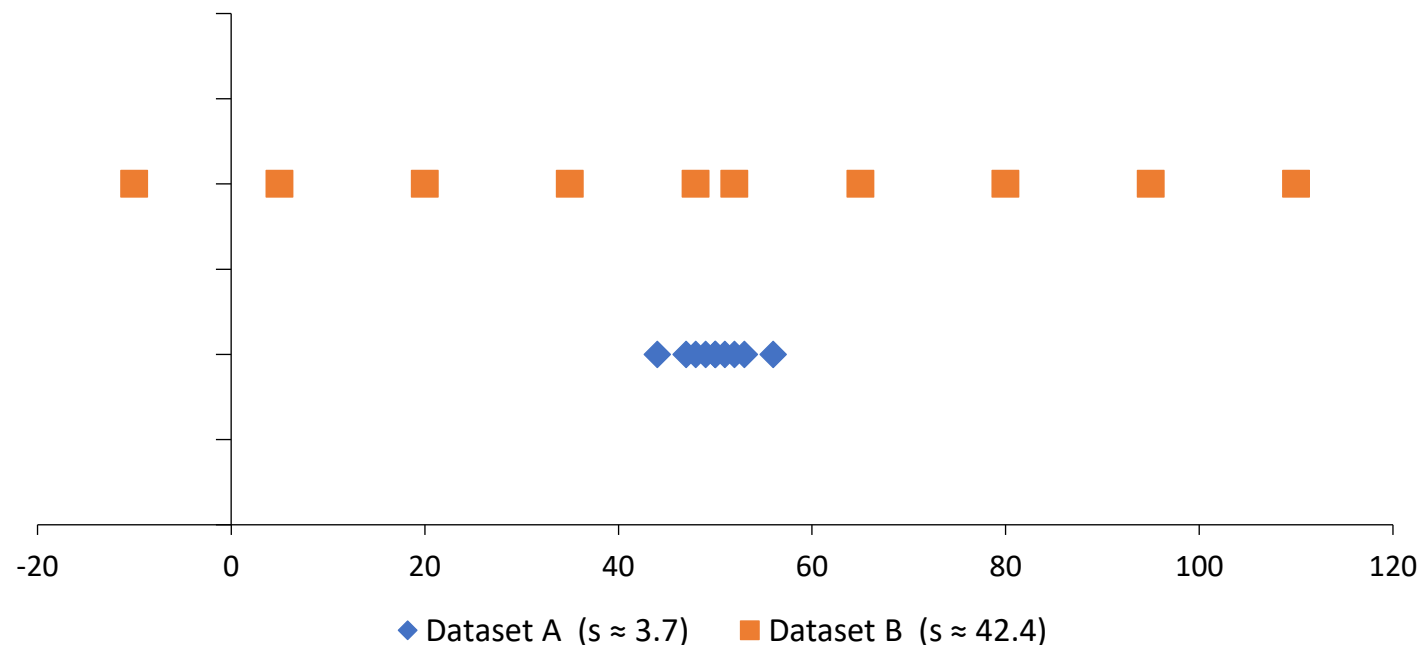
2

Measures of Dispersion

You will learn to quantify how widely data spreads, and to report the spread that fits the data.

Why Spread Matters

Two datasets, the same mean, completely different behaviour



Same centre

Both datasets have a mean of 50. A centre alone cannot distinguish them.

Different risk

Dataset A is predictable; Dataset B ranges from -10 to 110. The same decision is safe in one and reckless in the other.

Dispersion tells you how much the centre can be trusted — how wrong that one number is allowed to be.

Four standard measures: range, variance, standard deviation and the interquartile range (IQR).

Range, Variance and Standard Deviation

Measuring the average distance from the mean

Range

Largest value minus smallest. Easy to compute, but based on only two observations and grows with sample size.

$$R = x_{\text{max}} - x_{\text{min}}$$

Variance s^2

Mean of the squared deviations from $\bar{x}$. Squaring stops positive and negative deviations cancelling — but the units become “marks squared”.

$$s^2 = \Sigma(x_i - \bar{x})^2 / (n - 1)$$

Standard deviation s

Square root of the variance, so it returns to the original units. This is the figure you quote beside the mean.

$$s = \sqrt{s^2}$$

Speaking marks (n = 5)

Range = 22

$s^2 = 69.2$ marks²

$s = 8.32$ marks

Report it as

context only

for calculation only

“mean 37.2 ± 8.3 marks”

Divide by $n - 1$, not n : it corrects the bias that arises when a sample is used to estimate a population.

Percentiles, Quartiles and the IQR

The spread you quote next to a median

Definition

The pth percentile is the value below which p% of the observations fall.

$$\text{IQR} = \text{Q3} - \text{Q1}$$

The width of the middle 50% of the data

Statistic	Meaning
Q1 (25th pct)	25% of values lie below it
Q2 (50th pct)	The median
Q3 (75th pct)	75% of values lie below it
IQR	Spread of the central half

Quote the pair that matches the shape: mean ± s for symmetric data, median + IQR for skewed data.

Worked example — 2,400 orders

Q1	Median	Q3	IQR
142	268	497	355

Reading the asymmetry

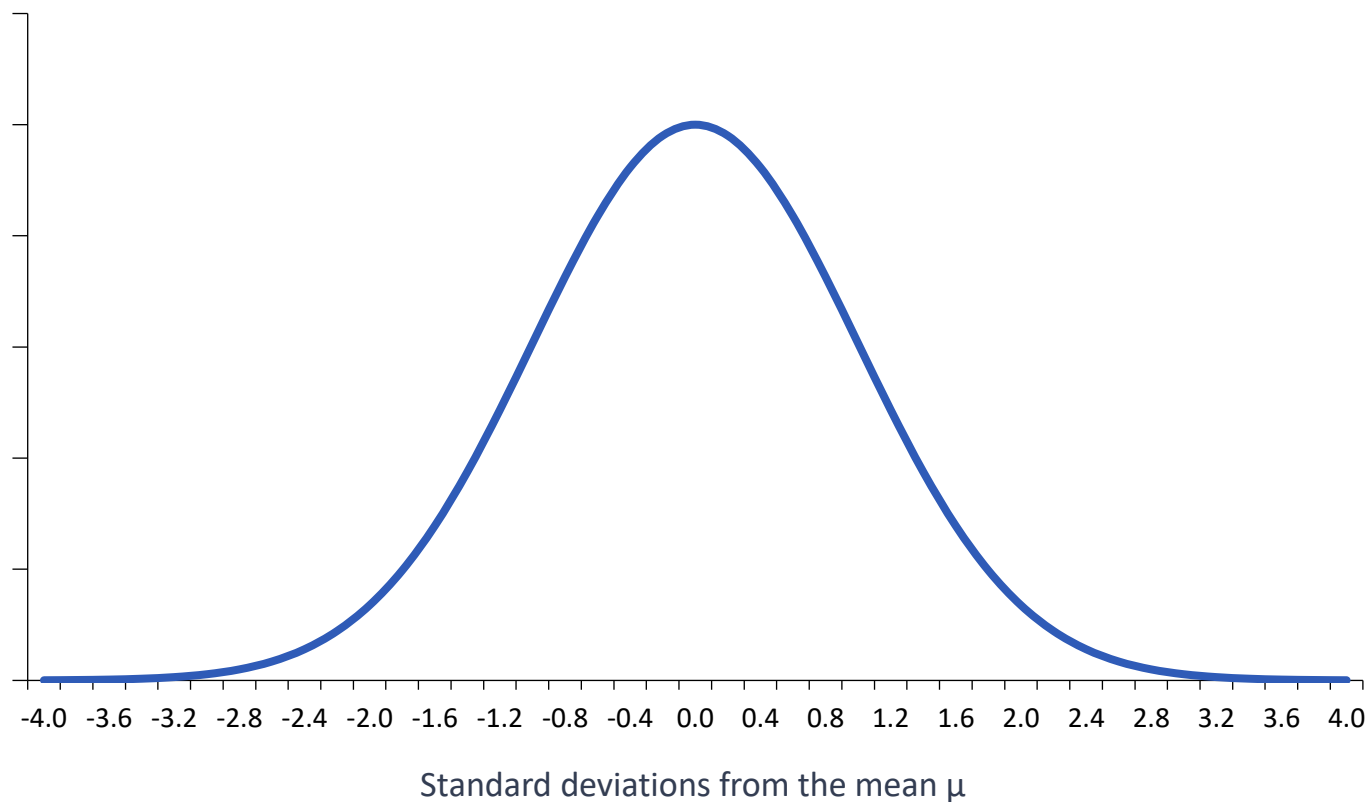
Q3 sits 229 above the median while Q1 sits only 126 below it — the data has a longer right tail.

Why the IQR is robust

It ignores the extreme quarter at each end, so single extreme values cannot inflate it.

The Normal Distribution and the 68–95–99.7 Rule

How the standard deviation partitions symmetric data



$\mu \pm 1s \rightarrow 68.3\%$

About two thirds of all observations.

$\mu \pm 2s \rightarrow 95.4\%$

The conventional boundary of “ordinary”.

$\mu \pm 3s \rightarrow 99.7\%$

Beyond this is 3 observations in 1,000.

The rule belongs to the normal curve, not to the standard deviation — test it before assuming it.

z-Scores: Standardising a Value

Converting “how far from the mean?” into a comparable number

$$z = (x - \bar{x}) / s$$

The number of standard deviations a value sits from the mean

z	Interpretation
< 1	Ordinary — within the central 68%
1 – 2	Somewhat unusual
2 – 3	Unusual — worth inspecting
> 3	Extreme — screen as a candidate outlier

PYTHON

```
z = (x - x.mean()) / x.std()
(z.abs() > 3).sum()    # candidate outliers
```

What standardising buys you

z removes the units, so values on different scales become comparable: a mark of 23 out of 50 ($z = -1.70$) and an order of 9,840 QAR ($z = +18.4$) sit on one scale.

It is also the standard input scaling for most machine-learning models.

Where it breaks

$|z| > 3$ is a sound screen for **symmetric data** and a **poor one** for **skewed data, where it flags the entire healthy tail.**

On skewed columns use percentiles instead.

Coefficient of Variation

Comparing spread across different scales and units

$$CV = s / \bar{x}$$

Unitless, so it compares columns measured on different scales

CV	Reading
< 0.3	Consistent — mean and s describe the column well
0.3 – 1.0	Variable — report both centre and spread

> 1.0 Spread exceeds the average — stop using mean and s

Never compute the CV when the mean is near zero or can be negative (temperatures, dates, profit).

Worked comparison

Column	Mean	s	CV
Speaking /50	37.2	8.32	0.22
Math /50	40.2	5.85	0.15
Order value QAR	418.60	512.30	1.22

Reading the three rows

Speaking marks are about 40% less consistent than math marks, even though both are out of 50.

Order value scores CV = 1.22: “419 ± 512” would imply negative orders, so switch to median and IQR.

Percentiles in Practice

Averages describe the middle; commitments live in the tail

Business question	Statistic	Value	Why this one
What does a typical order look like?	Median	3 days / 268 QAR	Describes one unit
What delivery time can we promise?	p95	7 days	Must survive the bad cases
How much capacity must we build?	p99	11 days of load	Sized for peak, not average
What revenue will n orders produce?	Mean × n	418.60 × n	Only the mean multiplies up

The average is the wrong promise

Mean delivery is 3.4 days. Advertise “3-day delivery” and roughly 45% of orders arrive late, because an average sits in the middle of the outcomes.

The percentile is a keepable one

“95% of orders within 7 days” is honest, measurable, and needs no assumption about the shape of the distribution.

Habit: report p50, p90 and p95 together — three numbers describe a skewed column better than mean ± s.

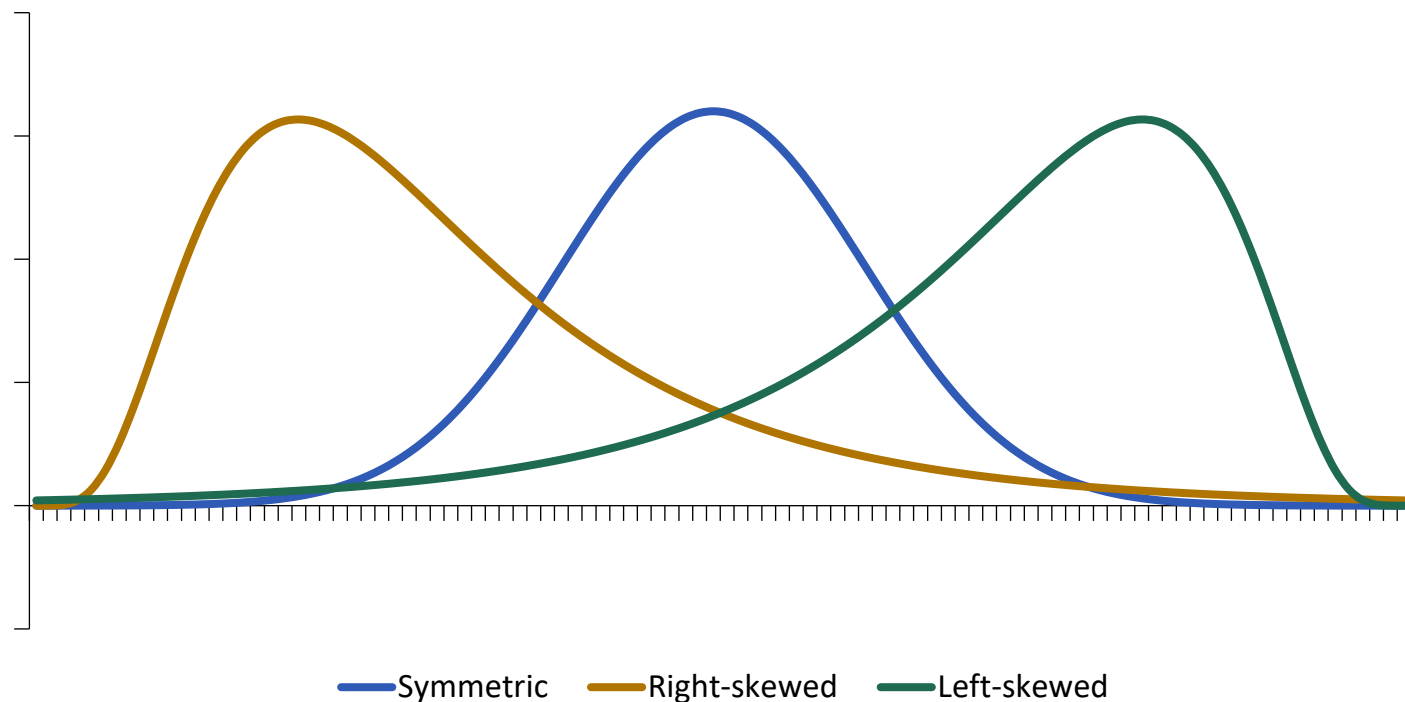
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Distribution Shape: Skewness and Kurtosis

You will learn to read the shape of a distribution and decide which statistics remain valid.

Symmetric, Right-Skewed and Left-Skewed

Shape decides which measure of centre you may report



Symmetric

Mean = median = mode. Report the mean.

Right-skewed (positive)

Long right tail; mean > median. Typical of income and revenue.

Left-skewed (negative)

Long left tail; mean < median. Typical of easy exam marks.

The two-second check: mean above median → right tail; mean below median → left tail.

The Coefficient of Skewness

Quantifying asymmetry around the mean

$$g_1 = (1/n) \sum (x_i - \bar{x})^3 / s^3$$

Cubing preserves the sign, so the direction of the tail survives

Skewness	Shape	Report
< -0.5	Left tail	Median
-0.5 to +0.5	Roughly symmetric	Mean
> +0.5	Right tail	Median + p90

PYTHON

```
df['price'].skew()      # +4.12
```

Worked example

Order value: skew = +4.12 — strongly right-skewed, the normal state of revenue, income and claim size.

Speaking marks: skew = -1.70, consistent with mean 37.2 < median 39.0.

The standard remedy

A log transform pulls in a long right tail: `log1p(order_value)` reduces skew from 4.12 to 0.21 — near symmetric. Analyse on the log scale, report on the original one.

Caution: skewness is unstable below roughly $n = 50$ — treat a small-sample value as a direction, not a measurement.

Kurtosis: The Weight of the Tails

How often extreme values occur, compared with a normal curve

$$g_2 = (1/n) \Sigma (x_i - \bar{x})^4 / s^4 - 3$$

pandas returns excess kurtosis: 3 is already subtracted, so 0 means normal-like tails

$$g_2 \approx 0$$

Normal-like tails. The 68–95–99.7 rule is a reasonable approximation.

$$g_2 > 0 \text{ (heavy tails)}$$

Extreme values occur more often than a normal curve predicts. Order value scores 26.8 — extreme orders are routine, not rare.

$$g_2 < 0 \text{ (light tails)}$$

Extreme values are rarer than normal; the data is concentrated near the centre.

Common misconception: kurtosis measures the peak. It does not — it measures the tails.

Practical consequence: with heavy tails, risk, capacity and service levels must be planned from percentiles, not from s.

Detecting Outliers

An outlier is a value far from the rest — two standard screens

Screen 1 — z-score rule (symmetric data)

About 99.7% of normally distributed values lie within $\bar{x} \pm 3s$.

Screen 2 — IQR fences (skewed data)

Based on quartiles, so no assumption of symmetry is required.

Outlier if $|x - \bar{x}| > 3s$

$x < Q1 - 1.5 \cdot IQR$ or $x > Q3 + 1.5 \cdot IQR$

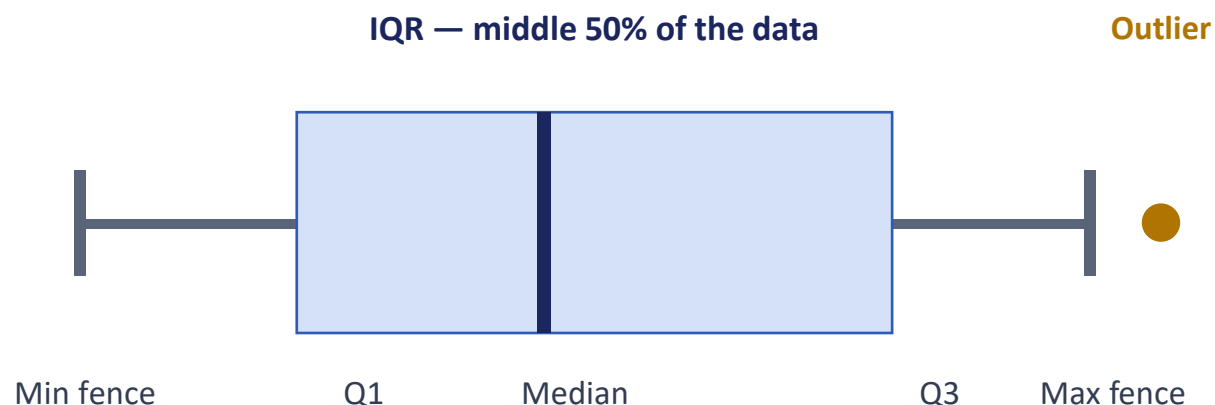
Orders data	Q1 = 142	Q3 = 497	IQR = 355	Upper fence = 1,029.50
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Result	118 of 2,400 flagged (4.9%)	Normal data would give $\approx 0.7\%$	The excess measures skew	It is not evidence of 118 errors
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A fence flags values that are far from the middle — not values that are wrong. Flagging is a question, not a verdict.

Boxplots: Reading Spread and Outliers at a Glance

One chart showing centre, spread, skew and extremes together



Whiskers extend to the furthest value within $1.5 \times \text{IQR}$ of the box; points beyond them are plotted individually.

What to read first

Where the median line sits inside the box. Off-centre means the distribution is skewed.

Why it beats a table

Centre, spread, asymmetry and extreme values are visible in one glance — and several groups can be compared side by side.

Box height = IQR • line inside the box = median • whiskers = $1.5 \times \text{IQR}$ • separate points = flagged outliers

In pandas: `df.boxplot(column='price', by='region')` – one call to compare the distribution across groups.

Handling Outliers: A Five-Step Procedure

Flagging a value is the beginning of the work, not the end

1

FLAG

Apply more than one screen: $1.5 \times \text{IQR}$ fences, $|z| > 3$, and domain limits such as a negative price.

2

TRACE

Open the source record. Who entered it, when, through which system? The row usually explains the value.

3

CLASSIFY

Impossible $\rightarrow$ fix or drop. Typing or unit error $\rightarrow$ correct. Genuine extreme $\rightarrow$ keep. Different population $\rightarrow$ split the analysis.

4

ACT

Correct, exclude with a note, keep and switch to robust statistics, or winsorise at p99. Decide by classification.

5

DOCUMENT

Report the figure with and without the excluded rows, and state the rule and the count.

Never do this

Never delete silently — an undisclosed removal that moves a reported figure is falsification.

Never trim data to make a test pass.

Never assume far = wrong

On the orders data the 118 flagged rows are the most valuable customers in the file. Remove them and every forecast is 14% low.

If removing the rows changes your conclusion, that change is itself the finding — report it.

4

Analysis of Relationships Between Variables

You will learn to measure how two variables move together — and what that measurement does not prove.

Covariance and Correlation

Direction alone, versus direction and strength

$$\text{cov}(X,Y) = \Sigma(x_i - \bar{x})(y_i - \bar{y}) / (n - 1)$$

$$r = \text{cov}(X,Y) / (s_x \cdot s_y)$$

Property	Covariance	Correlation r
Range	Unbounded	-1 to +1
Units	Product of the two units	Unitless
Tells you	Direction only	Direction and strength
Comparable across datasets?	No	Yes

Why correlation is readable

Correlation is covariance with the units divided out. A covariance of 1,184.6 means nothing on its own; $r = 0.31$ is immediately interpretable.

Square it for the honest size

$r^2 = 0.10$ — discount level explains 10% of the variation in order value. The remaining 90% is something else.

Pearson's Correlation Coefficient

The strength of a linear relationship between two continuous variables

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{[\sum (x - \bar{x})^2 \cdot \sum (y - \bar{y})^2]}}$$

r	Interpretation
+1.0	Perfect positive linear relationship
+0.3 to +0.7	Moderate positive
-0.3 to +0.3	Weak or none
-0.7 to -1.0	Strong negative

PYTHON

```
df[['price','sales']].corr()  
from scipy.stats import pearsonr
```

How to read the sign

Positive r: as one variable increases, the other tends to increase.
Negative r: as one increases, the other tends to decrease.

r = 0 means no LINEAR relationship — a curved relationship can still be strong.

Context sets the bar

These bands are a rule of thumb. In physics **r = 0.7 is weak**; in social research it is **remarkable**.

When Pearson's r Is Valid

Three assumptions — and what to do when they fail

1

Linearity

The relationship must follow a straight line. A curved pattern can give $r \approx 0$ even when the variables are strongly related.

2

Normality

Both variables should be approximately normally distributed, without heavy tails or extreme values.

3

Independence

Each observation must be independent of the others — no repeated measures of the same unit.

If this fails	Symptom	Do this instead
Linearity	Pearson and Spearman disagree sharply	Use Spearman, or transform the variable
Normality	Strong skew or outliers present	Use Spearman's rank correlation
Independence	Repeated or clustered observations	Model the structure; r is not valid

Always plot the scatter before reporting r — a single number cannot reveal a curve, a cluster or a reversal.

Spearman's Rank Correlation

A non-parametric measure based on ranks, not raw values

$$\rho = 1 - 6\sum d^2 / [n(n^2 - 1)]$$

d is the difference between the two ranks of an observation

Rank X	Rank Y	d	d ²
1	2	-1	1
2	4	-2	4
3	1	2	4
4	5	-1	1
5	3	2	4
		$\sum d^2$	14

Worked result

$$\rho = 1 - (6 \times 14) / [5 \times (25 - 1)] = 1 - 84/120 = 0.30$$

A value near 0: the two rankings agree only weakly.

Monotonic, not linear

Spearman asks only whether the variables move consistently in the same direction, at any rate.

Use it when

The data is skewed, ordinal, or contains outliers — exactly the cases where Pearson's assumptions fail.

Correlation Is Not Causation

A number between -1 and $+1$ never identifies which variable moved the other

Four rival explanations for any observed correlation between A and B

1

A causes B

Check that A reliably precedes B in time.

2

B causes A

Large orders may earn discounts, not the reverse.

3

A third variable drives both

Split the data by it — if the sign flips, you have found it.

4

Coincidence

20 columns give 190 pairs; at $\alpha = 0.05$ about 9 appear “significant” by chance.

Say it safely

Instead of “discounts drive larger orders”, write: “heavily discounted orders tended to be larger in Jan–Jun 2025 (Spearman $\rho = 0.46$, $n = 2,400$); discount explains about 10% of the variation, and the direction of cause is untested.”

Four phrases that keep you honest: “tended to”, “was associated with”, “in this period”, “in this sample”.

Section Recap: The Analysis Checklist

Work top to bottom on every new numeric column

#	Ask this	Use this	Then apply this rule
1	Is the column complete?	<code>isna().sum()</code> , <code>count</code>	Report n and the missing count first
2	What is typical?	<code>mean()</code> , <code>median()</code> , <code>mode()</code>	Gap > 10% of the median → report the median
3	How much spread?	<code>std()</code> , <code>quantile()</code> , IQR, CV	CV > 1 → use median and IQR, not mean ± s
4	What shape?	<code>skew()</code> , <code>kurtosis()</code>	skew > 1 → transform, or report percentiles
5	Any extreme values?	1.5 × IQR fence, z > 3	Flag, trace, classify, act, document
6	What may we promise?	<code>quantile([.9, .95, .99])</code>	Commitments use p95, never the mean
7	Do two columns relate?	<code>corr()</code> , <code>spearman</code> , r^2	Report r^2 and always attach a scatter plot
8	Does it survive a split?	<code>groupby().agg()</code>	If the direction flips, the subgroup table is the finding
9	What may we claim?	—	“Associated with”, plus n and period

Completeness → centre → spread → shape → extremes → relationship → claim.